Continuous From Snapshots to Streams: A Live State Graph Approach for LLM-Based Root-Cause-Analysis in Kubernetes

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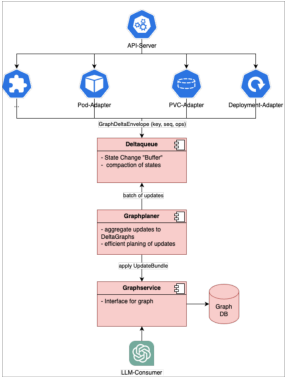

Outline

1 Introduction

2 Background

3 Research Question

Introduction

- Kubernetes and cloud-native technologies have become industry standard
- Key challenges: perceived complexity, lack of training [1]
- Root cause analysis (RCA) as key bottleneck

$\Rightarrow$ LLM-based assistance systems promise cognitive reduction and timely detection

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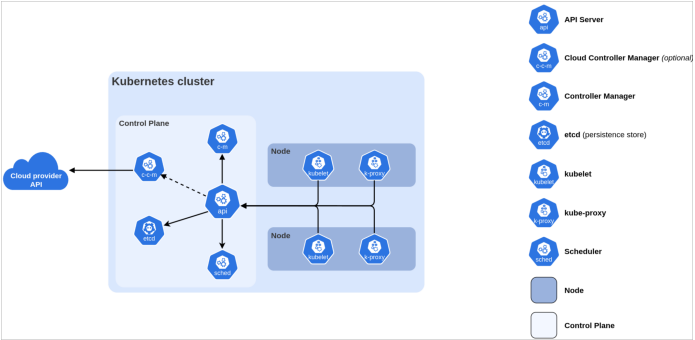
Kubernetes:

- Open-source, feature-rich container orchestrator
- Deployments through declarative manifests



Background

Error sources:

- Unforeseen events: node dies, network issues, ...
- Syntactically correct but semantically incorrect manifests

Error handling:

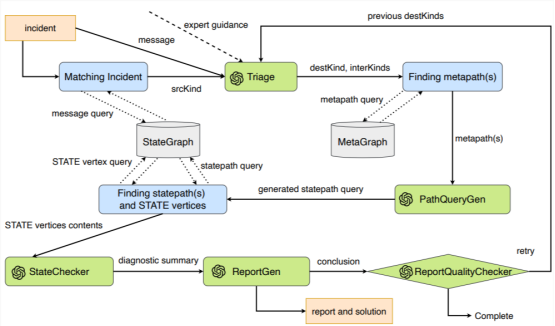
- Reading docs
- Manually debugging through API-resources with the CLI
- Applying fixes

```yaml
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: app-config
data:
  key: value
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 1
  selector:
    matchLabels:
      app: my-app
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
        - name: main
          image: nginx:1.27
          volumeMounts:
            - name: config
              mountPath: /etc/config
      volumes:
        - name: config
          configMap:
            name: app-config-typo
```

Background

Synergy-RCA [2]:

- periodically maps cluster snapshots to graph
- leverages LLMs to generate queries, checks, reports



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Research Question:

"Does a live state graph for Kubernetes and predefined traversal tools improve the quality, speed and interpretability under dynamic cluster conditions compared to snapshot-based approaches."

References I

Cloud Native Computing Foundation.
Cloud native 2024: Approaching a decade of code, cloud, and change, 2024.
[Online].

Yong Xiang, Charley Peter Chen, Liyi Zeng, Wei Yin, Xin Liu, Hu Li, and Wei Xu.
Simplifying root cause analysis in kubernetes with stategraph and llm, 2025.